Cyclic XOR

PLATFORM	DIFFICULTY	DATE
Platform: CodeChef	DIV 3	Solved: June 3, 2026

Problem

Apply

$$f(A)_i = A_{i-1} \oplus A_i \oplus A_{i+1}$$

exactly

$$2^K$$

times.

Count arrays that become all zeros.

Algorithm

Traverse string.

Find block length.

Add : ans += len / 3;

Example

```
aaaaaa
length = 6
6/3 = 2
Answer = 2
```

Key Observation

Define:

$$T = I + L + L^{-1}$$

where L is cyclic shift.

Over GF(2):

$$(a + b + c)^{2^K} = a^{2^K} + b^{2^K} + c^{2^K}$$

Thus:

$$T^{2^K} = I + L^{2^K} + L^{-2^K}$$

Formula

For every maximal block:

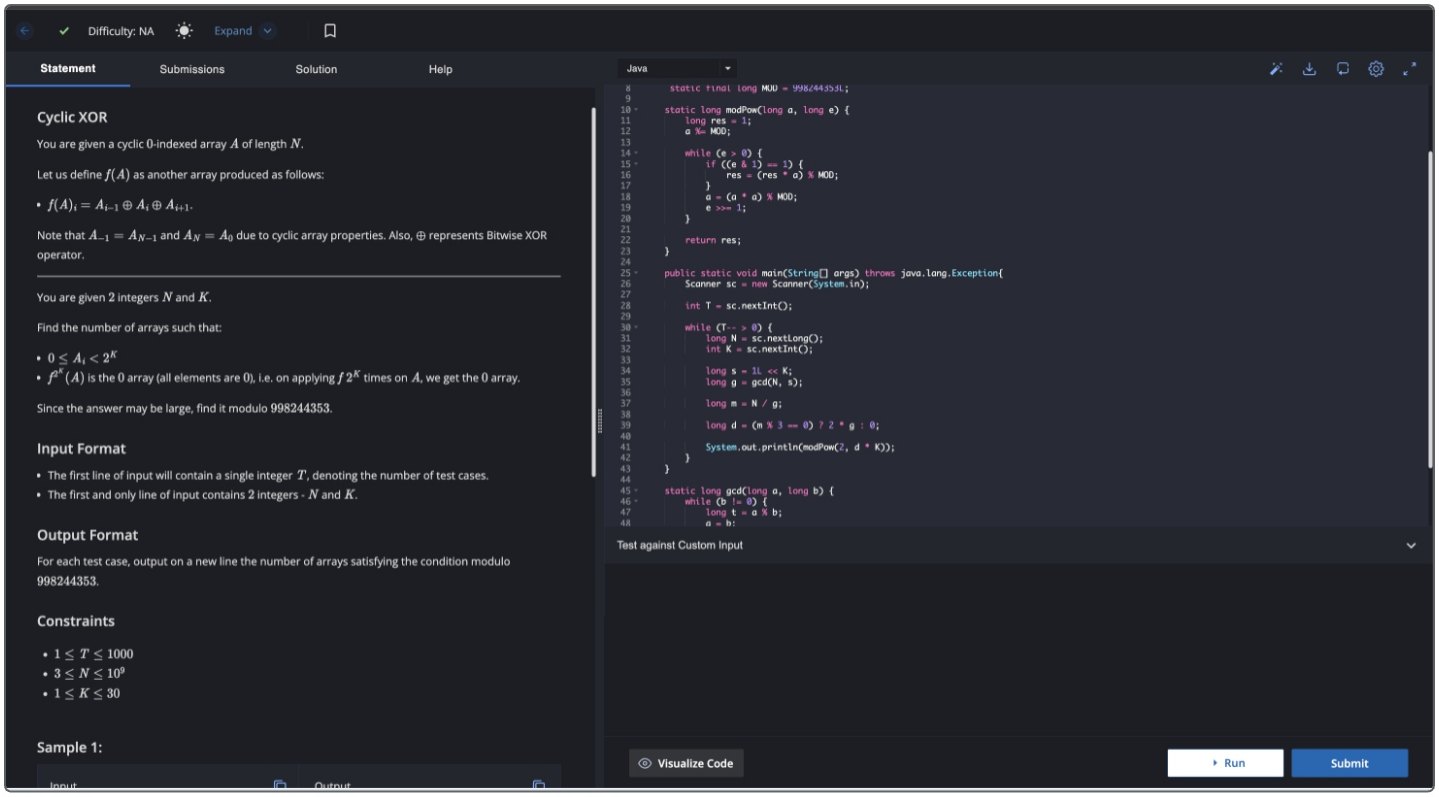
$$ans + = len/3$$

TC & SC

Time Complexity : O(log N)

Java Code

```
import java.util.*;
import java.lang.*;
import java.io.*;

class Codechef
{

         static final long MOD = 998244353L;

    static long modPow(long a, long e) {
        long res = 1;
        a %= MOD;

        while (e > 0) {
            if ((e & 1) == 1) {
                res = (res * a) % MOD;
            }
            a = (a * a) % MOD;
            e >>= 1;
        }

        return res;
    }

    public static void main(String[] args) throws ja
va.lang.Exception{
        Scanner sc = new Scanner(System.in);

        int T = sc.nextInt();

        while (T-- > 0) {
            long N = sc.nextLong();
            int K = sc.nextInt();

            long s = 1L << K;
            long g = gcd(N, s);

            long m = N / g;

            long d = (m % 3 == 0) ? 2 * g : 0;

            System.out.println(modPow(2, d * K));
        }
    }

    static long gcd(long a, long b) {
        while (b != 0) {
            long t = a % b;
            a = b;
            b = t;
        }
        return a;
    }
}
```


Difficulty: NA
Expand
Statement
Submissions
Solution
Help
Cyclic XOR
You are given a cyclic 0-indexed array A of length N.
Let us define $f(A)$ as another array produced as follows:
$f(A)_i = A_{i-1} \oplus A_i \oplus A_{i+1}$.
Note that $A_{-1} = A_{N-1}$ and $A_N = A_0$ due to cyclic array properties. Also, $\oplus$ represents Bitwise XOR operator.
You are given 2 integers N and K.
Find the number of arrays such that:
$0 \leq A_i < 2^K$
$f^{2^K}(A)$ is the 0 array (all elements are 0), i.e. on applying f 2^K times on A, we get the 0 array.
Since the answer may be large, find it modulo 998244353.
Input Format
The first line of input will contain a single integer T, denoting the number of test cases.
The first and only line of input contains 2 integers - N and K.
Output Format
For each test case, output on a new line the number of arrays satisfying the condition modulo 998244353.
Constraints
$1 \leq T \leq 1000$
$3 \leq N \leq 10^9$
$1 \leq K \leq 30$
Sample 1:
Input
Output
Java
static final long MOD = 998244353L;
static long modPow(long a, long e) {
long res = 1;
a %= MOD;
while (e > 0) {
if ((e & 1) == 1) {
res = (res * a) % MOD;
}
a = (a * a) % MOD;
e >>= 1;
}
return res;
}
public static void main(String[] args) throws java.lang.Exception{
Scanner sc = new Scanner(System.in);
int T = sc.nextInt();
while (T-- > 0) {
long N = sc.nextLong();
int K = sc.nextInt();
long s = 1L << K;
long g = gcd(N, s);
long m = N / g;
long d = (m % 3 == 0) ? 2 * g : 0;
System.out.println(modPow(2, d * K));
}
}
static long gcd(long a, long b) {
while (b != 0) {
long t = a % b;
a = b;
Test against Custom Input
Visualize Code
Run
Submit